

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week-3

2520030098

Gutha Sushma Sree

Section – 3

The screenshot displays the Databricks Data Explorer interface for a project named 'Project 0' under the organization 'GUTHA SUSHMA SRE...'. The interface is divided into a left sidebar and a main content area.

Left Sidebar (Data Explorer):

- My Queries:** Empty list.
- Data Modeling:** Empty list.
- CLUSTERS (1):** Cluster0 is selected.
- Search clusters:** Search bar.
- Cluster0:**
 - BankDB
 - admin
 - bookflow
 - book_metadata** (selected)
 - books
 - local
 - sample_mflix

Main Content Area (book_metadata):

Cluster0 > bookflow > book_metadata

Documents 5 | Aggregations | Schema | Indexes 1 | Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA | **BULK** | **EXPORT CODE** | 25 | 1 - 5 of 5

book_metadata

	_id ObjectId	book_id Int32	title String	author String	published_year Int32
1	ObjectId('6a7eb72ccd16c4c...')	101	"The Martian"	"Andy Weir"	2021
2	ObjectId('6a7eb759cd16c4c...')	102	"Dune"	"Frank Herbert"	2022
3	ObjectId('6a7eb76ccd16c4c...')	103	"Clean Code"	"Robert C. Martin"	2019
4	ObjectId('6a7eb77fcd16c4c...')	104	"Artificial Intelligence ..."	"John Smith"	2023
5	ObjectId('6a7eb795cd16c4c...')	105	"Data Structures"	"Mark Allen"	2024

book_metadata

	category String	format String	file_size String	member_id Int32	reviews Array
1	"Science Fiction"	"Digital PDF"	"2.5 MB"	101	[3 elements]
2	"Science Fiction"	"Digital EPUB"	"3.1 MB"	102	[3 elements]
3	"Programming"	"Physical"	No field	103	[2 elements]
4	"Artificial Intelligence"	"Digital PDF"	"4.2 MB"	104	[3 elements]
5	"Computer Science"	"Digital PDF"	"5.0 MB"	105	[2 elements]

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PROJECT
Project 0

Compare tiers

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- BankDB
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- bookflow
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book_metadata

Cluster0 > bookflow > book_metadata

View monitoringVisualize Your Data

Documents 5AggregationsSchemaIndexes 1Validation

Type a query: { field: 'value' } or [Generate query](#)

ExplainResetFindOptions

ADD DATA +BULKEXPORT CODE

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book_metadata

ing	file_size String	member_id Int32	reviews Array	dimensions Object
1 >F"	"2.5 MB"	101	[] 3 elements	No field
2 >UB"	"3.1 MB"	102	[] 3 elements	No field
3	No field	103	[] 2 elements	{ } 3 fields
4 >F"	"4.2 MB"	104	[] 3 elements	No field
5 >F"	"5.0 MB"	105	[] 2 elements	No field

Comparison operator \$gt

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book_metadata

Cluster0 > bookflow > book_metadata

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```
{  "reviews.rating": {    "$gt": 4  } }
```

Generate queryExplainResetFindOptions

ADD DATA +BULKEXPORT CODE

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book_metadata

_id ObjectID	book_id Int32	title String	author String	published_year Int32
1 ObjectId('6a7eb72ccd16c4c...	101	"The Martian"	"Andy Weir"	2021
2 ObjectId('6a7eb759cd16c4c...	102	"Dune"	"Frank Herbert"	2022
3 ObjectId('6a7eb76ccd16c4c...	103	"Clean Code"	"Robert C. Martin"	2019
4 ObjectId('6a7eb77fcd16c4c...	104	"Artificial Intelligence ..."	"John Smith"	2023
5 ObjectId('6a7eb795cd16c4c...	105	"Data Structures"	"Mark Allen"	2024

Using \$in

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book_metadata

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```
{  "reviews.rating": {    "$in": [4, 5]  } }
```

Generate queryExplainResetFindOptions

ADD DATA +BULKEXPORT CODE

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book_metadata

_id ObjectID	book_id Int32	title String	author String	published_year Int32
1 ObjectId('6a7eb72ccd16c4c...	101	"The Martian"	"Andy Weir"	2021
2 ObjectId('6a7eb759cd16c4c...	102	"Dune"	"Frank Herbert"	2022
3 ObjectId('6a7eb76ccd16c4c...	103	"Clean Code"	"Robert C. Martin"	2019
4 ObjectId('6a7eb77fcd16c4c...	104	"Artificial Intelligence ..."	"John Smith"	2023
5 ObjectId('6a7eb795cd16c4c...	105	"Data Structures"	"Mark Allen"	2024

Regex query

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Compare tiers

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book_metadata

Cluster0 > bookflow > book_metadata

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Generate query

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

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{}

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book_metadata

	_id ObjectId	book_id Int32	title String	author String	published_year
1	ObjectId('6a7eb72ccd16c4c...')	101	"The Martian"	"Andy Weir"	2021
2	ObjectId('6a7eb759cd16c4c...')	102	"Dune"	"Frank Herbert"	2022
3	ObjectId('6a7eb77fcd16c4c...')	104	"Artificial Intelligence ..."	"John Smith"	2023
4	ObjectId('6a7eb795cd16c4c...')	105	"Data Structures"	"Mark Allen"	2024

Aggregation pipeline

\$match

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\$match

Edit

Explain

Run

Options

ALL RESULTS

EXPORT CODE

Showing 1 - 4 count results

☰

{}

🏠

book_metadata

_id: ObjectId('6a7eb72ccd16c4c1e1e67084')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

reviews: Array (3)

_id: ObjectId('6a7eb759cd16c4c1e1e67087')

book_id: 102

title: "Dune"

author: "Frank Herbert"

published_year: 2022

category: "Science Fiction"

format: "Digital EPUB"

file_size: "3.1 MB"

member_id: 102

reviews: Array (3)

\$unwind

Data Explorer

- My Queries
- Data Modeling

CLUSTERS (1)

Search clusters

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Cluster0 > bookflow > book_metadata

View monitoring Visualize Your Data

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Sunwind Edit

Explain Run Options

ALL RESULTS EXPORT CODE

Showing 1 – 13 count results < > ⌵ ☰

```
_id: ObjectId('6a7eb72ccd16c4c1e1e67084')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
reviews: Object (3)
```

```
_id: ObjectId('6a7eb72ccd16c4c1e1e67084')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
reviews: Object (3)
```

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Sunwind Edit

Explain Run Options

ALL RESULTS EXPORT CODE

Showing 1 – 13 count results < > ⌵ ☰ ⓘ

```
_id: ObjectId('6a7eb72ccd16c4c1e1e67084')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
reviews: Object (3)
```

```
_id: ObjectId('6a7eb759cd16c4c1e1e67087')
book_id: 102
title: "Dune"
author: "Frank Herbert"
published_year: 2022
category: "Science Fiction"
format: "Digital EPUB"
file_size: "3.1 MB"
member_id: 102
reviews: Object (3)
```

Alert System Status: All Good

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ALL RESULTS

EXPORT CODE

Showing 1 - 13 count results

_id: ObjectId('6a7eb759cd16c4c1e67087')

book_id: 102

title: "Dune"

author: "Frank Herbert"

published_year: 2022

category: "Science Fiction"

format: "Digital EPUB"

file_size: "3.1 MB"

member_id: 102

reviews: Object (3)

_id: ObjectId('6a7eb759cd16c4c1e67087')

book_id: 102

title: "Dune"

author: "Frank Herbert"

published_year: 2022

category: "Science Fiction"

format: "Digital EPUB"

file_size: "3.1 MB"

member_id: 102

reviews: Object (3)

Alert System Status: All Good

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\$group

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ALL RESULTS

EXPORT CODE

Showing 1 - 5 count results

_id: "Artificial Intelligence Basics"

average_rating: 4.333333333333333

total_reviews: 3

_id: "The Martian"

average_rating: 4.666666666666667

total_reviews: 3

_id: "Clean Code"

average_rating: 4.5

total_reviews: 2

_id: "Dune"

average_rating: 4.666666666666667

total_reviews: 3

_id: "Data Structures"

average_rating: 4

total_reviews: 2

Alert System Status: All Good

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\$sort

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\$sort

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ALL RESULTS

EXPORT CODE

Showing 1 - 5 count results

_id: "Dune"

average_rating: 4.666666666666667

total_reviews: 3

_id: "The Martian"

average_rating: 4.666666666666667

total_reviews: 3

_id: "Clean Code"

average_rating: 4.5

total_reviews: 2

_id: "Artificial Intelligence Basics"

average_rating: 4.333333333333333

total_reviews: 3

_id: "Data Structures"

average_rating: 4

total_reviews: 2

Alert System Status: All Good

Create text index

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Validation

Create Index

Refresh

Looking for search indexes?

These indexes can be created and viewed under [Search and Vector Search](#).

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR	36.9 kB	3 (since Tue Aug 25 2024)	UNIQUE	READY
> reviews.comments_text	TEXT	20.5 kB	0 (since Fri Aug 28 2026)		READY

Alert System Status: All Good

Search using text index

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 - book_metadata**
 - books
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 - sample_mflix

book_metadata

Cluster0 > bookflow > book_metadata

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```
{  "$text": {    "$search": "excellent"  }}
```

Generate query + Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1 - 5 of 5

	_id ObjectId	book_id Int32	title String	author String	published_year Int32	
1	ObjectId('6a7eb72ccd16c4c...	101	"The Martian"	"Andy Weir"	2021	
2	ObjectId('6a7eb76ccd16c4c...	103	"Clean Code"	"Robert C. Martin"	2019	
3	ObjectId('6a7eb759cd16c4c...	102	"Dune"	"Frank Herbert"	2022	
4	ObjectId('6a7eb795cd16c4c...	105	"Data Structures"	"Mark Allen"	2024	
5	ObjectId('6a7eb77fcd16c4c...	104	"Artificial Intelligence ...	"John Smith"	2023	

Search for specific keyword

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CLUSTERS (1)

Search clusters

- Cluster0
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 - admin
 - bookflow
 - book_metadata**
 - books
 - local
 - sample_mflix

book_metadata

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

```
{  "$text": {    "$search": "\"science fiction\""}  }
```

Generate query + Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1 - 2 of 2

	_id ObjectId	book_id Int32	title String	author String	published_year Int32	
1	ObjectId('6a7eb759cd16c4c...	102	"Dune"	"Frank Herbert"	2022	
2	ObjectId('6a7eb72ccd16c4c...	101	"The Martian"	"Andy Weir"	2021	

Use explain()

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CLUSTERS (1)

Search clusters

- Cluster0

book_metadata

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

```
{  "$text": {    "$search": "excellent"  }}
```

Generate query + Explain Reset Find Options

Output:

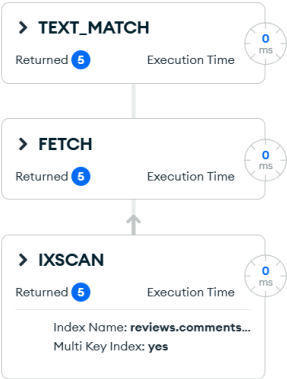
Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

Interpret ⓘ

Visual Tree Raw Output

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Query Performance Summary

- 5 documents returned
 - 5 documents examined
 - 0 ms execution time
 - Is not sorted in memory
 - 5 index keys examined
- Query used the following index:
- _fts (text)
 - _ftsx ↑

Close